

JNTU Hyderabad

REGULAR EXAMINATION ? 2026

Power Systems-II ? Regular Examination, 2026

Subject Code: EE501 Semester: 5 Regulation: R20

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Fault Analysis)

1. Define Fault Analysis. Explain its significance in Power Systems-II with a suitable example. [12 Marks]
2. Derive the expression related to Fault Analysis and explain each term. [12 Marks]
3. Derive the expression related to Fault Analysis and explain each term. [12 Marks]

SECTION B (Unit 2: Power System Stability)

4. How is Power System Stability implemented? Give an algorithm / procedure and explain. [8 Marks]
5. Define Power System Stability. Explain its significance in Power Systems-II with a suitable example. [12 Marks]
6. Compare and contrast the key concepts involved in Power System Stability. [12 Marks]

SECTION C (Unit 3: Economic Load Dispatch)

7. Describe the working principle of Economic Load Dispatch with necessary equations. [8 Marks]
8. Define Economic Load Dispatch. Explain its significance in Power Systems-II with a suitable example. [12 Marks]
9. Define Economic Load Dispatch. Explain its significance in Power Systems-II with a suitable example. [12 Marks]

SECTION D (Unit 4: HVDC and FACTS)

10. Derive the expression related to HVDC and FACTS and explain each term. [8 Marks]
11. Explain HVDC and FACTS with a neat diagram/sketch and a suitable example. [8 Marks]
12. List the advantages and limitations of HVDC and FACTS. Explain any two in detail. [12 Marks]

SECTION E (Unit 5: Power System Transients)

13. Define Power System Transients. Explain its significance in Power Systems-II with a suitable example. [8 Marks]
14. Discuss the engineering applications of Power System Transients. [12 Marks]

15. Describe the working principle of Power System Transients with necessary equations.

[12 Marks]

*** End of Question Paper ***